Ruoyu Wen

PhD Candidate in Product Design - Human-Computer Interaction

Human-Agent Interaction · Game Design · Virtual Reality · Generative AI

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Google Scholar · LinkedIn · Portfolio

Profile

HCI researcher and game designer working on **human-agent interaction**, **virtual reality**, **alternate reality games**, and **generative AI**. Author of **15+ peer-reviewed publications** (CHI, IUI, CHI Play, DiGRA, SIGGRAPH Asia), with international research experience across New Zealand, Germany, Korea, Sweden, and China.

Research Interests: Human-Agent Interaction · Virtual & Extended Reality · Alternate Reality Games · Player Psychology · Generative AI · Cross-cultural HCI

Languages: Chinese (Native) · English (Fluent)

Education

University of Canterbury - Christchurch, New Zealand

PhD, School of Product Design · 2023 - 2027 (expected Nov 2027)

Human-agent interaction, VR, and player motivation in ARGs through Self-Determination Theory; mixed-methods experimental research.

Advisors: Assoc. Prof. Thammathip Piumsomboon, Assoc. Prof. Simon Hoermann, Prof. Mark Billinghurst.

Uppsala University - Uppsala, Sweden

Master of Arts in Game Design · 2021 - 2023

Interactive narrative, player psychology, user experience, and game design theory.

Advisor: Assoc. Prof. Sarah Lynne Bowen.

Huazhong University of Science and Technology - Wuhan, China

Double Bachelor's Degree - Communication Studies (first major) & Information Science (second major) · 2016 - 2021

Digital media, social media, human-computer interaction, and digital culture.

Advisor: Assoc. Prof. Shuo Xiong (熊硕).

Honors and Grants

Research Grants in Germany - DAAD (Deutscher Akademischer Austauschdienst) · 2026

Competitive research grant supporting a research stay at the University of Augsburg, Chair for Human-Centered AI.

Chinese Government Award for Outstanding Self-Financed Students Abroad - China Scholarship Council (CSC) · 2025

National award recognising outstanding self-funded PhD students overseas (University of Canterbury, New Zealand).

Research and Professional Experience

Visiting PhD Student - University of Augsburg, Chair for Human-Centered AI (Prof. Elisabeth André), Augsburg, Germany

Apr 2026 - Jul 2026

• Collaborated on human-centered AI, intelligent user interfaces, and human-agent interaction.

Lead AI Game Designer - MICROFEEL, Remote

Oct 2024 - Jul 2026

• Led a cross-functional team building an end-to-end system that generates game stories, mechanics, and NPCs from natural-language input.

• Translated generative-AI research into shipping product features.

Visiting PhD Student - KAIST, GSCT UVR Lab, Daejeon, South Korea

Oct 2025

• Collaborative research on extended reality (XR) and human-computer interaction.

Teaching Assistant - University of Canterbury, Christchurch, New Zealand

Jul 2025 - Oct 2025

• TA for *Computation for Games*; mentored undergraduates in game development, programming, and computational foundations.

PhD Researcher - University of Canterbury, School of Product Design, Christchurch, New Zealand

2023 - Present

• Designed and ran VR and human-agent interaction studies on conversational AI, including experiment design, data collection, and mixed-methods analysis.

• First-author and co-author publications at CHI, OzCHI, DiGRA, and SIGGRAPH Asia.

Research Assistant - University of Canterbury, Christchurch, New Zealand

Jul 2024 - Jan 2025 · Part-time

• Supported HCI and VR studies through experimental design, data collection, and analysis.

Research Intern - NeXT SCENE (Immersive Entertainment), Shanghai, China

Apr 2022 - May 2022

• Studied narrative-driven entertainment formats (script murder, escape rooms) and player engagement in social gaming.

Strategy Investor Intern - Huya (NASDAQ: HUYA), Guangzhou, China

Feb 2021 - Jun 2021

• Market research and investment analysis across gaming and entertainment technology.

Game Designer - Chengdu ChuangLeHui (JOY NOVA), Sichuan, China

Mar 2020 - May 2020

• Concept design, gameplay design, and playtesting for the mobile game *Champion of the Fields*.

Selected Publications

For the complete list, see Google Scholar.

1. **AI of Oz: Enhancing Wizard of Oz Studies in HCI with AI Assistance for Human Moderation.** *ACM CHI*, 2026.

2. **From Prompt to Presence: Co-Creating Personalised Emotional Sanctuaries in VR with Generative AI.** *ACM IUI*, 2026.

3. **Can AI Prompt Humans? Multimodal Agents Prompt Players' Game Actions and Show Consequences to Raise Sustainability Awareness.** *ACM CHI*, 2025.

4. **Exploring User Preferences for Museum Guides: The Role of Chatbots in Shaping Interactive Experiences.** *CHI Extended Abstracts*, 2025.

5. **GenLinguaScape: Enabling User-Defined VR Scenarios for Communicative Language Practice.** *IEEE ISMAR Adjunct*, 2025.

6. **Large Language Models for Automatic Detection of Sensitive Topics.** *OzCHI*, 2024.

7. **An Exploratory Study on AI-driven Visualisation Techniques for Decision Making in Extended Reality.** *OzCHI*, 2024.

8. **Sketchar: Supporting Character Design and Illustration Prototyping Using Generative AI.** *Proc. ACM HCI (CHI Play)*, 2024.

9. **Motivational Landscapes of ARG Players: A Self-Determination Theory Perspective.** *DiGRA*, 2024.

10. **Virtual Triplets: Mixed-Modal Synchronous and Asynchronous Collaboration with Human-Agent Interaction in VR.** *CHI Extended Abstracts*, 2024.

11. **I Light U Up: Exploring a New Emergent Narrative Paradigm through Light and Generative AI.** *SIGGRAPH Asia Art Papers*, 2024.

12. **Listen to the Sword: Using Breathing and Spatial Audio for Wuxia Games.** *CHI Extended Abstracts*, 2024.

13. **The Impact of Alternate Reality Games on Environmental Cognition for University Freshmen.** *HCI International*, 2024.

14. **Player Category Research on Murder Mystery Games.** *International Journal of Role-Playing*, 2023.

15. **The Chinese Hotpot of Larp.** *Knutpunkt Magazine*, 2022.